BioSuite-NG

User Manual

Radiobiological treatment-optimisation software

Developed by: To keep the author's name anonymous for the reviewers, the researcher's name has been removed in this version.

Based on the methodology of: Uzan J, Nahum AE. Radiobiologically guided optimisation of the prescription dose and fractionation scheme in radiotherapy using BioSuite. *Br J Radiol.* 2012;85(1017):1279–1286. doi:10.1259/bjr/20476567

DISCLAIMER

IMPORTANT NOTES FOR USERS OF BIOSUITE-NG

- Neither the developer nor Ahvaz Jundishapur University of Medical Sciences (AJUMS) can be held responsible for any issue that may arise from using this software for the treatment of patients.
- This program is **not destined for clinical use**.
- The final clinical decision should always lie with the clinician in charge of the case.

BioSuite-NG is a research/educational reimplementation and extension of the BioSuite software described by Uzan & Nahum (2012). All NTCP/TCP model parameters shown or shipped as defaults come from published literature fits to specific patient cohorts and carry the uncertainty inherent to those fits — this is discussed further in the Confidence Intervals section of this manual.

Contents

1. Installation and launching the app

Option A — one-click launcher (recommended)

- Double-click **run_biosuitepy.bat**. The first run installs all required Python packages automatically (needs Python 3.10+ already installed on Windows); subsequent runs open faster.

Option B — standalone .exe (no Python needed afterwards)

- Run **build_exe.bat** once. This produces **dist\BioSuite-NG.exe**, which you can copy anywhere and double-click directly — no terminal, no Python install required on the target machine.
- If startup speed matters more than having a single file, run **build_exe_fast_start.bat** instead (produces a **dist\BioSuite-NG** folder that starts faster on every launch — copy the whole folder, not just the .exe inside it).

Option C — manual command line

- `pip install -r requirements.txt`
- `python main.py`

Why does it take a moment to open?

BioSuite-NG imports several large libraries at startup (PyQt6, matplotlib, scipy, pandas, and optionally pydicom) — typically 2-5 seconds with a normal Python install. A --onefile .exe is slower still on its first launch because Windows has to unpack the bundled interpreter to a temporary folder before anything can run. To make sure you always get instant feedback that the app IS launching (and hasn't silently failed), a splash screen with the AJUMS logo appears immediately on double-click, before any of those slow imports happen.

2. Overall workflow

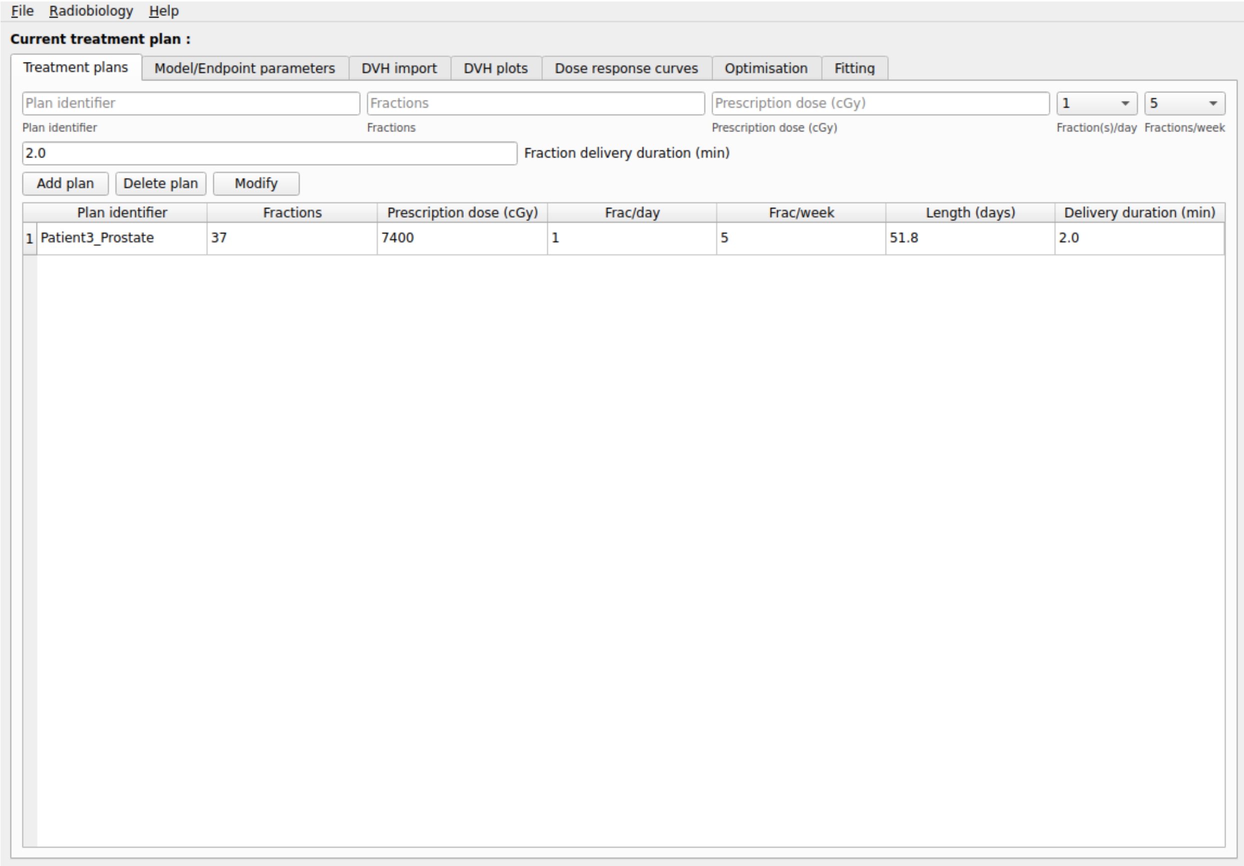
BioSuite-NG follows the same one-active-plan-at-a-time model as the original BioSuite: everything you do across every tab refers to the **current treatment plan** shown at the top of the window. The recommended order of work is:

- **Treatment plans** — create/select the plan (fractions, prescription dose, etc).
- **Model/Endpoint parameters** — define which NTCP/TCP models and parameters you will use (or load a saved/default bank).
- **DVH import** — load each structure's DVH and associate it with one of the endpoints you just defined.
- **DVH plots** — visually check the DVHs and read off TCP/NTCP, confidence intervals, and sensitivity.
- **Dose response curves** / **Optimisation** — explore dose escalation and find the isotoxic optimum.
- **Fitting** — (optional) calibrate the TCP model's α against your own cohort's clinical outcomes.

Note this is why **Model/Endpoint parameters** comes before **DVH import** in the tab order: you need at least one endpoint defined before a DVH can usefully be associated with it.

3. Treatment plans tab

This is where you create and manage treatment plans. A plan defines the prescription (fractions, dose, schedule) that every other tab will use for its calculations.



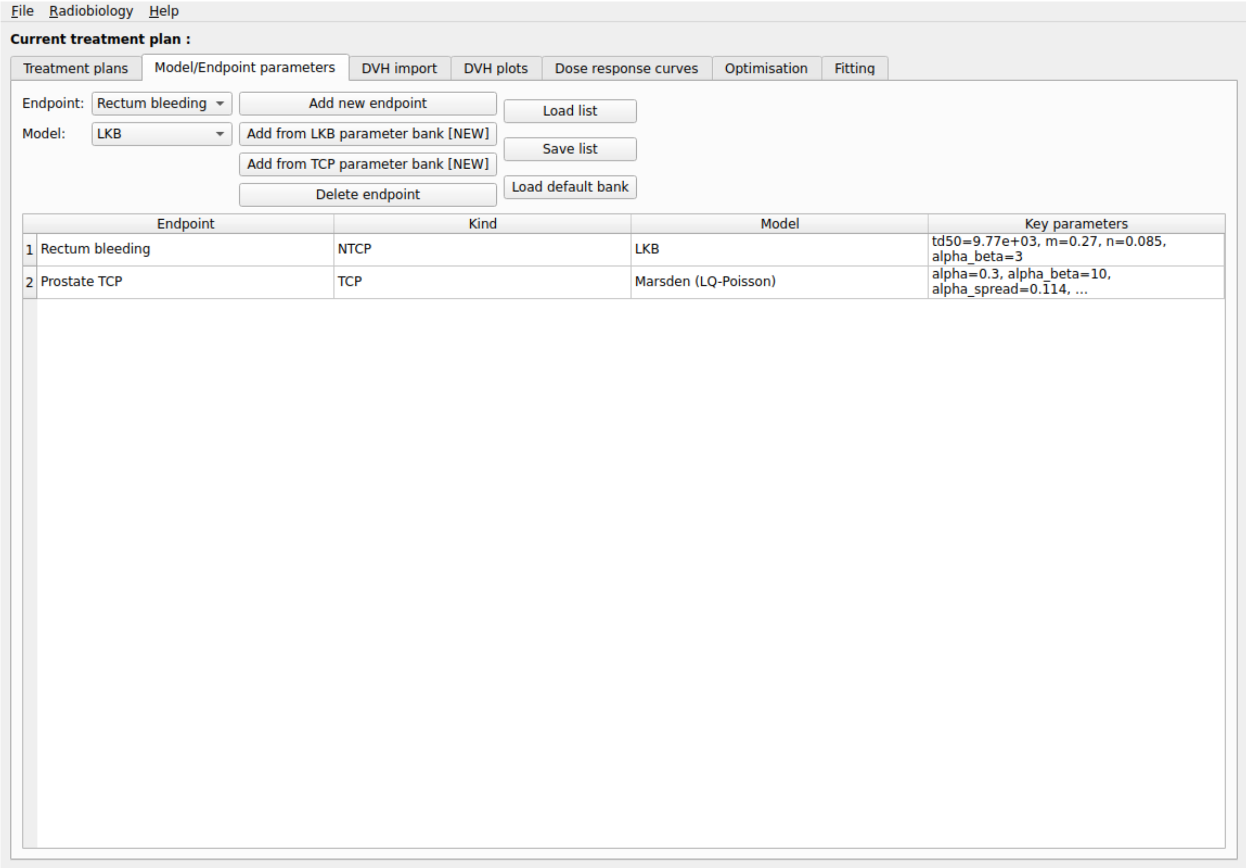
The treatment plans tab in BioSuite-NG.

What each part does:

- **Plan identifier** — a name you choose for this plan (e.g. patient ID or 'Patient3_Prostate').
- **Fractions** — total number of fractions.
- **Prescription dose (cGy)** — total physical dose.
- **Fraction(s)/day** and **Fractions/week** — schedule, used to compute overall treatment time (relevant for repopulation in the TCP model).
- **Add plan / Modify / Delete plan** — standard CRUD buttons; click a row in the table first to Modify or Delete it.

4. Model/Endpoint parameters tab

An **endpoint** pairs a clinical outcome (e.g. 'Rectum bleeding', 'Prostate TCP') with a radiobiological model and its parameters. You define endpoints here BEFORE associating DVHs with them in the next tab.



The model/endpoint parameters tab in BioSuite-NG.

What each part does:

- **Add new endpoint** — opens a dialog where you choose NTCP or TCP, pick a model (LKB / Relative Seriality / SMD / EUD for NTCP; Marsden or LQ-SLR for TCP), and enter its parameters.
- **Add from LKB parameter bank [NEW]** / **Add from TCP parameter bank [NEW]** — pick a literature-sourced parameter set instead of typing values by hand; see the next section for both.
- **Delete endpoint** — click a row in the table to select it, THEN click Delete endpoint; it always removes whichever row is actually highlighted.
- **Load default bank** — loads `data/default_endpoints.json`, a starter set of organs (Lung, Rectum, Oesophagus) with literature parameters from the original paper's Table 1. Replace this file with your own parameter bank at any time.
- **Load list / Save list** — import/export your own set of endpoints as a JSON file, so you don't have to re-enter them every session.

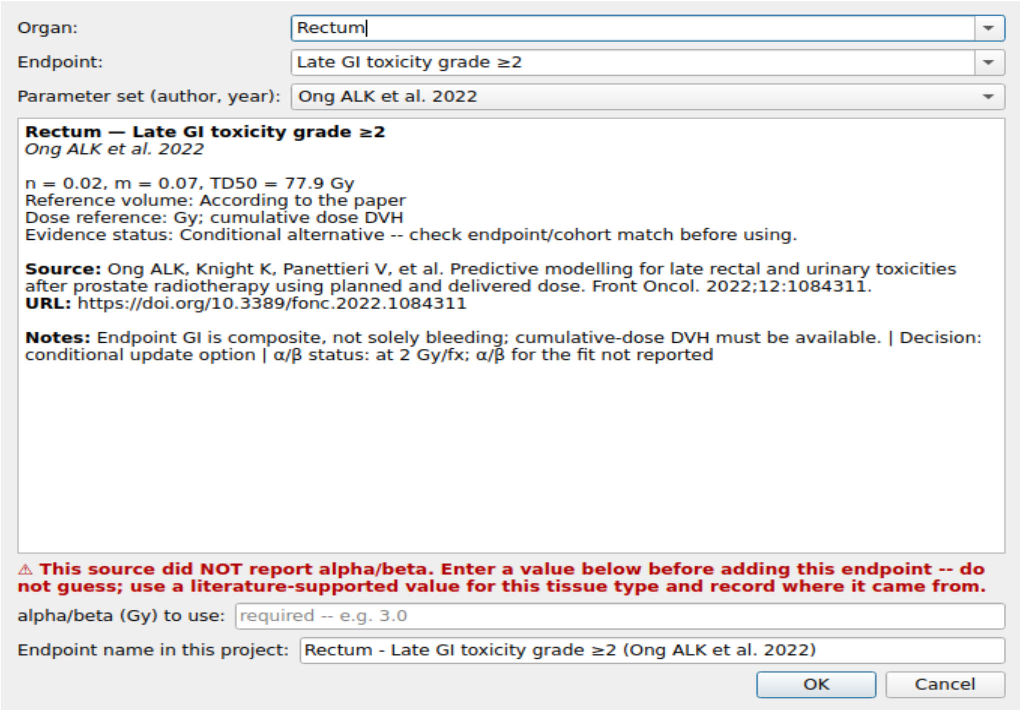
Tips:

- The summary table shows every parameter for every endpoint at a glance — use it to double check values before running calculations.

5. Adding endpoints from the LKB and TCP parameter banks

LKB parameter bank (NTCP)

Instead of typing LKB parameters in by hand, click **'Add from LKB parameter bank [NEW]'** in the Model/Endpoint parameters tab. This opens a curated, literature-sourced bank of **88 parameter sets across 51 organs**, built from a structured evidence review (classic Burman et al. 1991 values, QUANTEC-era updates, and newer conditional alternatives with full citations).



The 'Add from LKB parameter bank' dialog.

What each part does:

- **Organ** → **Endpoint** → **Parameter set (author, year)** — three cascading dropdowns, all searchable-as-you-type [NEW] (type e.g. "rect" to jump straight to Rectum, case-insensitive, matches anywhere in the name). Some organ/endpoint combinations have more than one citable parameter set (e.g. a historical Burman 1991 value AND a newer conditional alternative) — pick whichever matches your case by its “Author et al. Year” label.
- **Detail panel** — shows n, m, TD50, the reference volume, the dose reference (physical / EQD2 / EQD25 / Gy(RBE) — these are NOT interchangeable), the evidence status, and the full citation.
- **alpha/beta warning** — the LKB model itself only has three parameters (TD50, m, n); alpha/beta is only needed for fractionation (EQD2) correction. When the source paper did not report one, BioSuite-NG shows a red warning and **requires** you to type a value in before the endpoint can be added — it is never silently guessed for you.

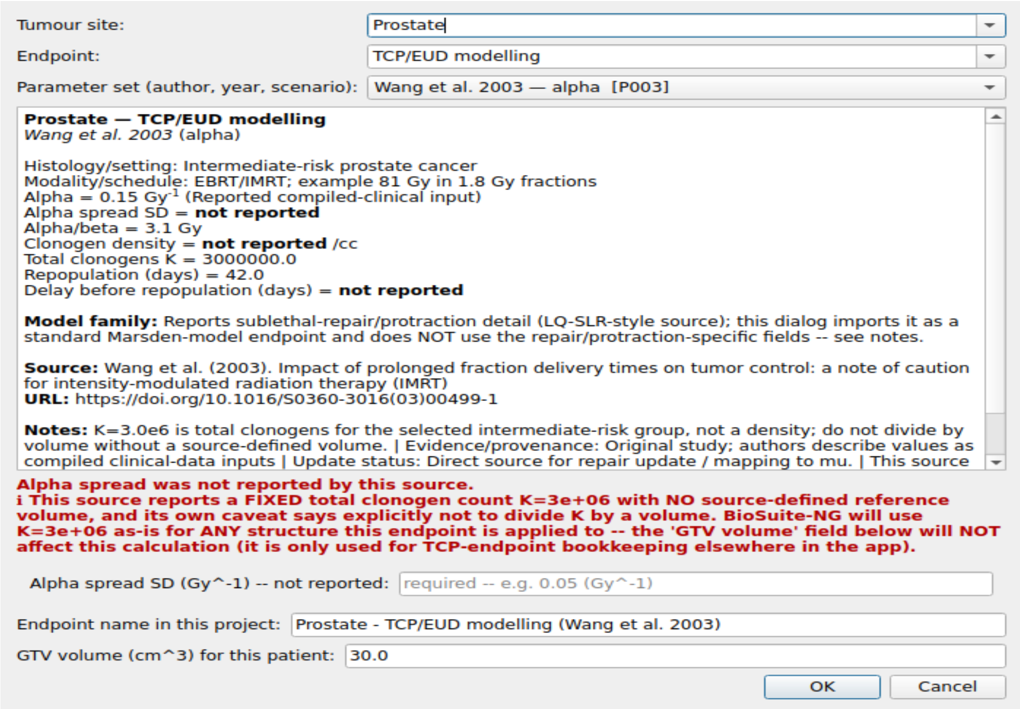
Tips:

- Open **Radiobiology** → **'LKB parameter bank (methodology & references)'** at any time for the full methodology, the meaning of every evidence-status label, and the primary references — this is what lets you double-check a parameter set before trusting it.

- A 2023 systematic review found over 500 different published LKB parameter sets with substantial disagreement between them — always sanity-check a bank entry against your own institution's experience before using it to inform a real decision.

TCP parameter bank

The same idea, for tumour control: click **'Add from TCP parameter bank [NEW]'** in the Model/Endpoint parameters tab to pick from **14 Target/Poisson TCP parameter sets across 9 tumour sites** (lung, prostate, cervix, bladder, breast, CNS, liver, oesophagus, rectum).



The 'Add from TCP parameter bank' dialog.

What each part does:

- **Tumour site** → **Endpoint** → **Parameter set** — same cascading, searchable-as-you-type pattern as the LKB bank.
- **Model family** — only records BioSuite-NG's engine can actually reproduce are selectable: *computable_full* (matches the Marsden engine directly) or *alpha_beta_evidence_only* (the source reports ONLY alpha/alpha-beta -- you must supply density, spread and repopulation yourself). Records using an incompatible formalism (e.g. Zaider-Minerbo re-sensitisation) are documented in the bank but not offered here.
- **Total clonogens K vs. density — the single most important rule in this bank:** some sources report an absolute clonogen COUNT (K) for their cohort, not a per-cc density, and explicitly state it must NOT be divided by an arbitrary volume. When that's the case, BioSuite-NG uses K exactly as reported, for ANY structure the endpoint is later applied to -- the 'GTV volume' field will NOT change the result for those records (this is intentional, not a bug: an earlier version of this dialog let K be divided by a typed-in volume, which silently contradicted the source's own caveat and could produce a misleadingly-small TCP).
- Any field the record doesn't report (alpha spread, density/K, repopulation timing) appears as a required input below the detail panel — nothing is silently defaulted.

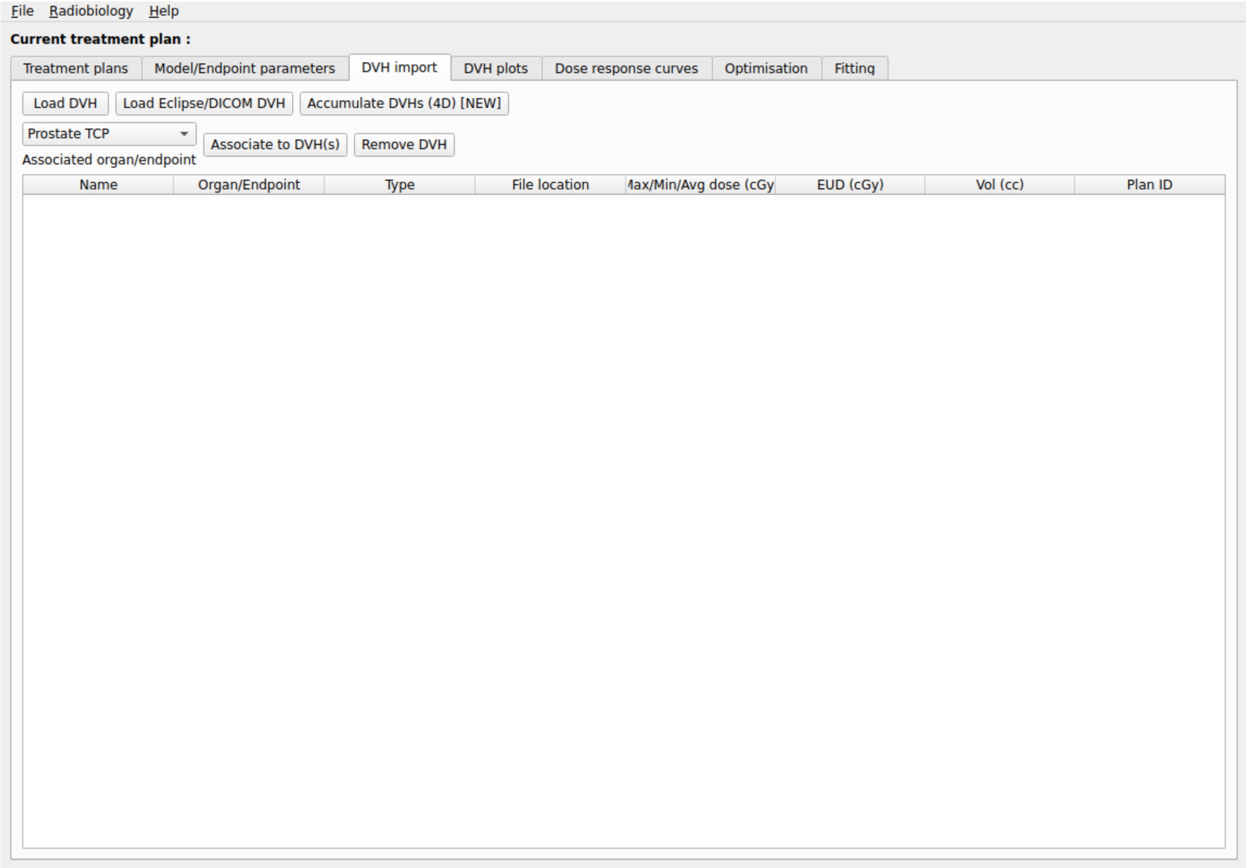
Tips:

- Open **Radiobiology** → **'TCP parameter bank (methodology & references)'** for the full methodology and the K-vs-density rule in detail.

- TCP models assume the DVH represents an actual tumour target receiving a full course of treatment. Applying a tumour's clonogen count to a normal-tissue/OAR DVH (or to a DVH with a cold spot, i.e. some volume at/near zero dose) will correctly predict TCP near 0% -- see the DVH plots tab's TCP tooltip for a live explanation when this happens.

6. DVH import tab

Load each structure's dose-volume histogram (DVH) here, and link (associate) it to one of the endpoints you defined in the previous tab.



The dvh import tab in BioSuite-NG.

What each part does:

- **Load DVH** — opens a file picker accepting Excel (native Pinnacle export format, or generic wide/long layout) or CSV files.
- **Load Eclipse/DICOM DVH** — imports directly from a DICOM RTDOSE + RTSTRUCT pair (no manual export needed).
- **Accumulate DVHs (4D) [NEW]** — select two or more DVHs of the SAME structure (e.g. different breathing phases, or primary course + boost) and combine them into one accumulated DVH.
- **Associated organ/endpoint + Associate to DVH(s)** — select one or more DVH rows, pick an endpoint from the dropdown, and click Associate. This link is what lets the DVH plots, Optimisation, and Dose response curves tabs compute TCP/NTCP.
- **EUD (cGy) column** — the generalised EUD (Niemierko formula) computed on the EQD2-corrected DVH, using the volume exponent implied by the associated endpoint's model. If a DVH is not yet associated with an endpoint, this column falls back to the simple mean dose and is marked with an asterisk (*).

Tips:

- You MUST associate a DVH with an endpoint before the DVH plots, Dose response curves, and Optimisation tabs will show any TCP/NTCP numbers for it.
- **Comparing the SAME structure under two different predictive parameter sets** (e.g. an OAR scored with both Author A's and Author B's LKB parameters): each endpoint-DVH link in BioSuite-NG is one

fixed pairing, so load the SAME DVH file a SECOND time (Load DVH again, same file) to get a second, independent row for that structure, then associate that second copy with the second endpoint. You now have two rows -- same physical DVH, two different models/parameter sets -- and can compare their NTCP/TCP side by side in DVH plots or Optimisation. Repeat once per additional parameter set you want to compare.

7. DVH plots tab

The main visual/numerical check tab: plot any combination of structures' DVHs, and read off TCP/NTCP (with optional confidence intervals and sensitivity analysis) for the currently-selected structure.

The dvh plots tab in BioSuite-NG.

What each part does:

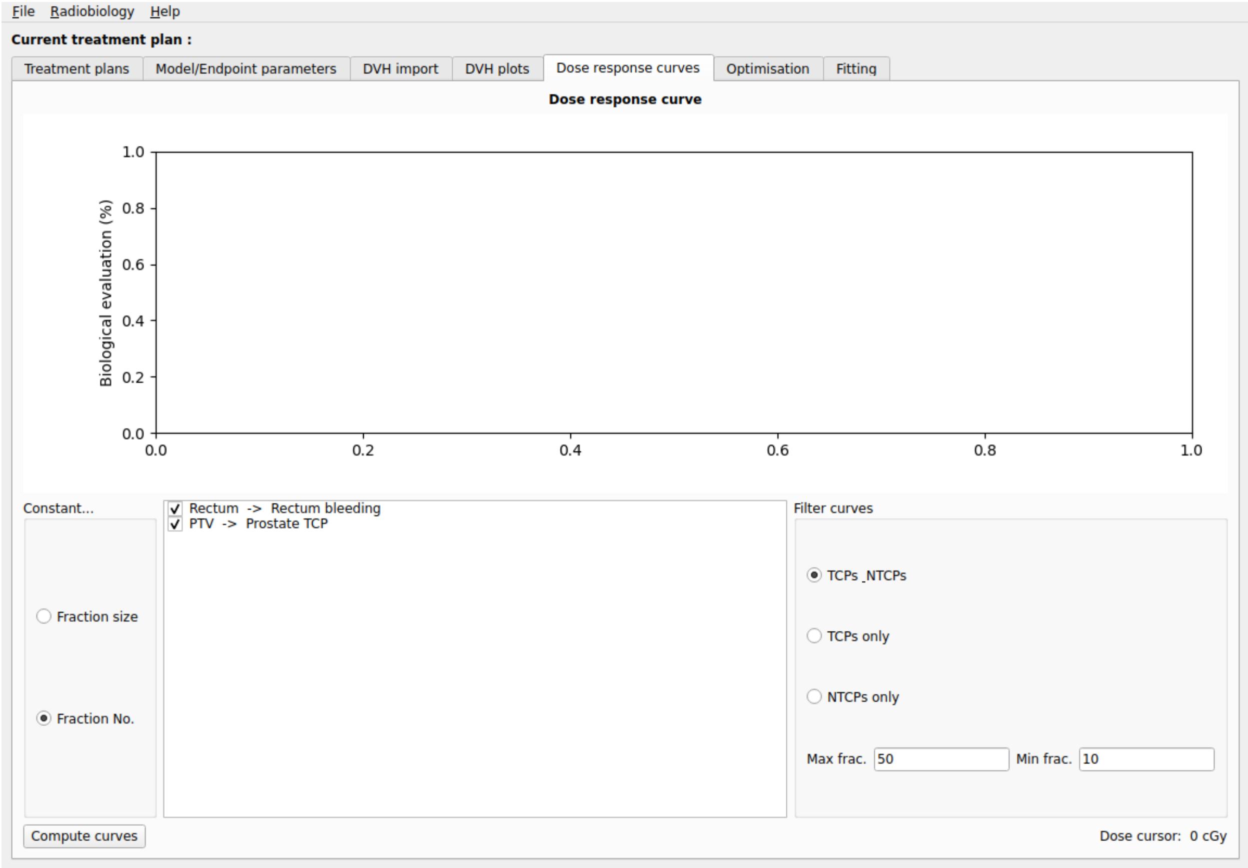
- **Structure list (checkboxes)** — tick/untick which DVHs are drawn on the chart; click a row to select it (this is what the TCP/NTCP boxes and CI/Tornado buttons act on).
- **TCP / LQ corrected NTCP boxes** — show the value for the currently-selected, endpoint-associated structure.
- **Compute 95% CI (Monte-Carlo) [NEW]** — propagates ±10% uncertainty on the model's key parameters through 1000s of simulated NTCP calculations, reporting the resulting 95% range. See Section 12 for how to interpret this.
- **Sensitivity (tornado) [NEW]** — opens a chart showing which single parameter drives the most uncertainty in NTCP.
- **Display options** — Normalised (chart title switches between 'Absolute value DVH(s)' and 'Normalized DVH(s)'), Differential/Cumulative DVH type, and Uncorrected/LQ-corrected dose display.

Tips:

- The chart uses a fine graph-paper-style grid specifically so you can read off approximate values directly from the curve.

8. Dose response curves tab

Reproduces Figures 1-2 of the original paper: shows how TCP/NTCP change as you escalate dose, either at a constant number of fractions (varying fraction size) or at a constant fraction size (varying the number of fractions).



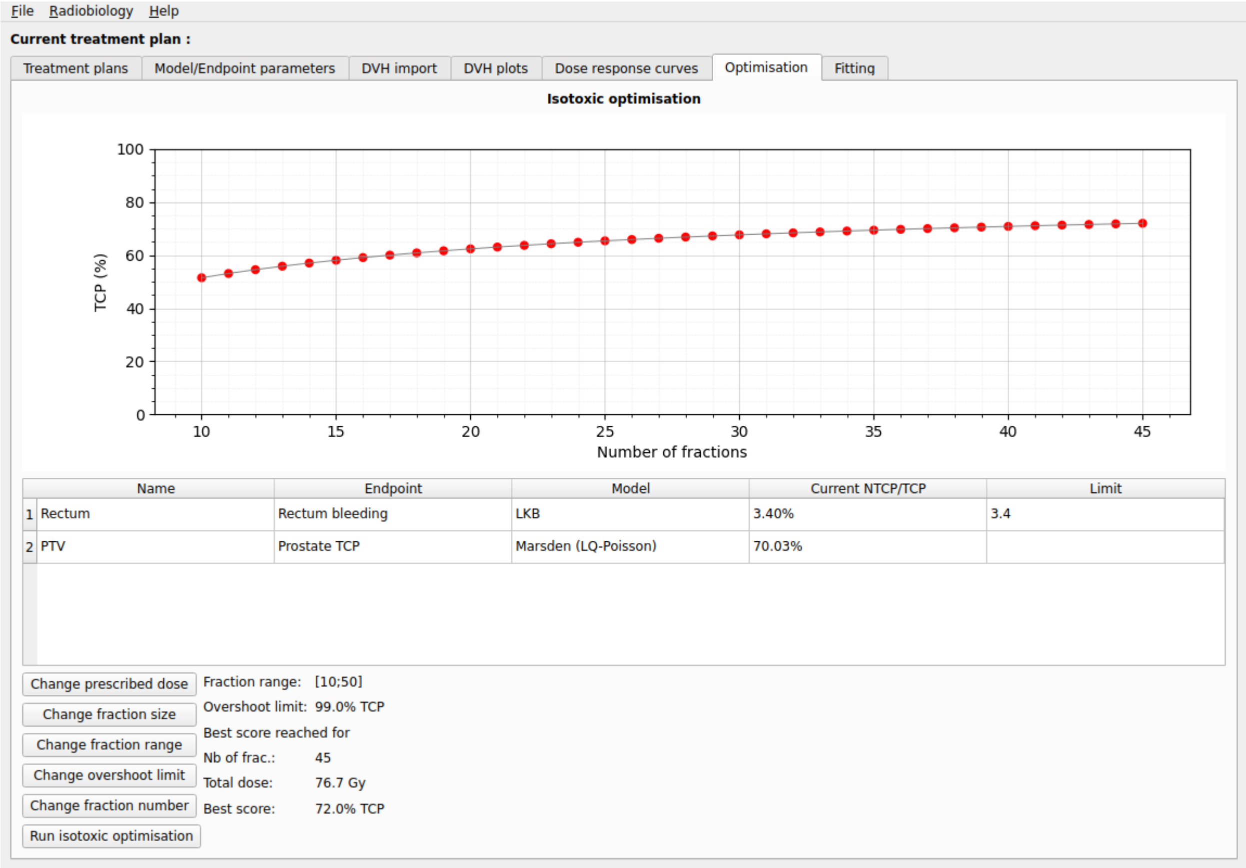
The dose response curves tab in BioSuite-NG.

What each part does:

- **Constant... Fraction size / Fraction No.** — choose which quantity stays fixed while the other is swept.
- **Structure/endpoint list** — tick which associated structures appear on the chart.
- **Filter curves** — show TCPs only, NTCPs only, or both.
- **Max/Min frac.** — the fraction-number range to sweep when 'Fraction size' is held constant.
- **Compute curves** — runs the sweep and redraws the chart.

9. Optimisation tab

Reproduces Figures 3-5 of the original paper: the isotoxic 2D optimisation finds, for each number of fractions in a range, the highest total dose that keeps every associated NTCP endpoint at or below its limit, and plots the resulting TCP.



The optimisation tab in BioSuite-NG.

What each part does:

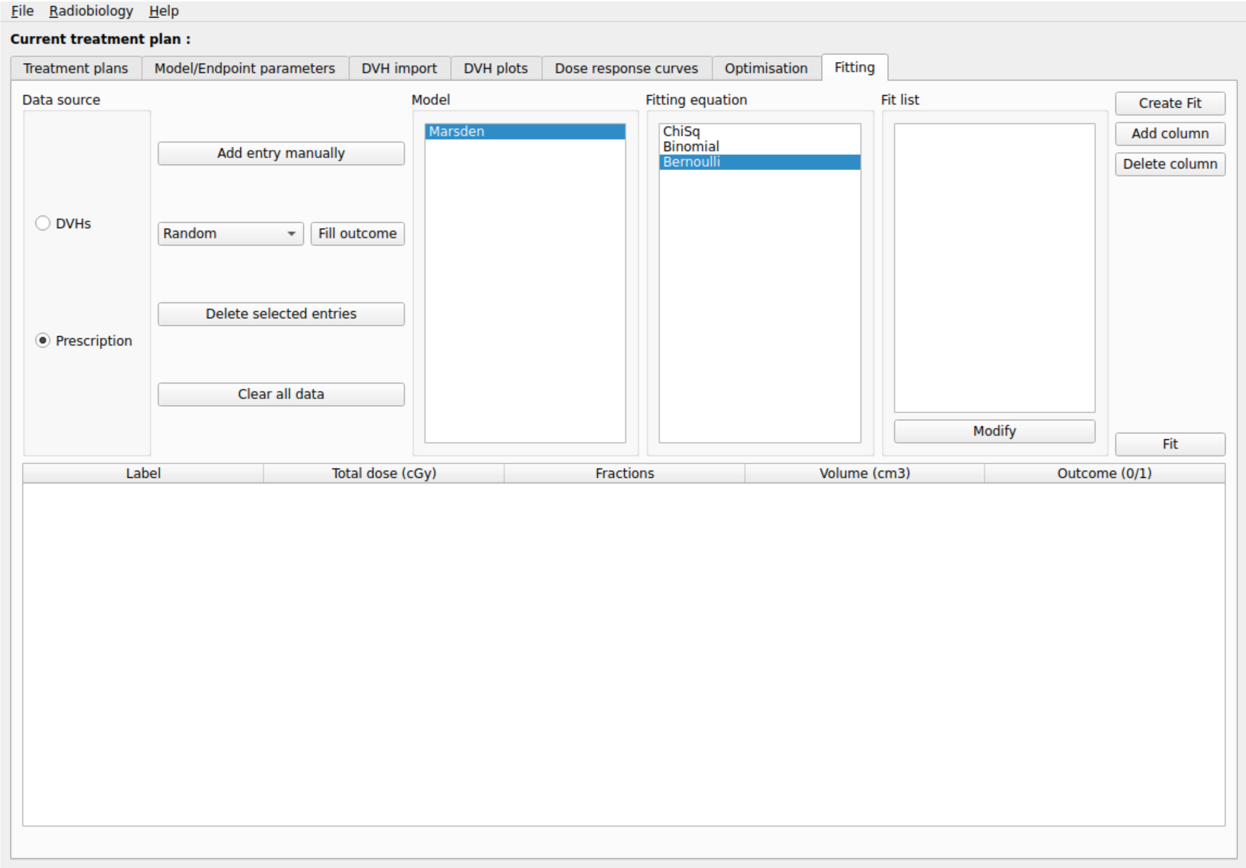
- **Table** — every associated endpoint, its current value, and (for NTCP endpoints) an editable **Limit** column — set this to the maximum acceptable NTCP for that organ.
- **Change fraction range / overshoot limit** — controls the fraction-number search range and the TCP ceiling above which the optimiser stops escalating dose (green dots), vs. being genuinely NTCP-limited (red dots).
- **Run isotoxic optimisation** — executes the search and reports the best fraction number / total dose / TCP combination.

Tips:

- You need exactly one TCP-kind endpoint associated for this tab to work; all NTCP-kind associated endpoints become simultaneous constraints.

10. Fitting tab

The opposite direction from Optimisation: given real clinical outcome data (local control yes/no per patient), this tab fits the TCP model's α parameter by maximum likelihood, so the model matches YOUR patient cohort rather than only literature defaults.



The fitting tab in BioSuite-NG.

What each part does:

- **Add entry manually** — enter one patient's total dose, fractions, GTV volume, and outcome (1 = local control, 0 = failure).
- **Fill outcome** — bulk-set all entries' outcomes (useful for quick testing).
- **Model** — currently Marsden (LQ-Poisson) only.
- **Fitting equation** — ChiSq / Binomial / Bernoulli objective function for the fit.
- **Fit** — runs the fit and reports the fitted α and the resulting objective-function score in the Fit list.

Tips:

- Only α is fitted; α/β, α-spread and clonogen density are held fixed at the values in Model/Endpoint parameters. This matches how the original paper only varies α/tumour α/β as a scenario, not a per-cohort fit.

11. Export

Reachable from **Radiobiology** → **Export...** at any time (not tied to a specific tab), this dialog gets data OUT of BioSuite-NG in plain CSV/JSON files you can open in Excel or any other tool.

- **NTCP/TCP summary for the active plan (CSV)** — one row per associated structure, with its endpoint, model, and computed NTCP/TCP value.
- **Raw DVH data for the active plan's structures (CSV)** — every dose bin of every loaded structure, long-format (Structure, Dose_cGy, Volume_cm3).
- **Endpoint list (JSON)** — identical format to Model/Endpoint parameters' 'Save list', so you can archive or share exactly which endpoints/parameters were used for a given analysis.

12. New features not in the original BioSuite

BioSuite-NG adds several capabilities the original 2012 BioSuite did not have. The table below summarises where each lives and what it's for.

Feature	Where	What it's for
EUD/Niemierko NTCP model	Model/Endpoint parameters (Model = 'EUD')	An additional NTCP model option beyond LKB/RS/SMD.
Monte-Carlo 95% CI	DVH plots → 'Compute 95% CI' button	Shows how much NTCP could plausibly vary given ±10% uncertainty on the model's own parameters. Wired up for LKB, RS, SMD and EUD models.
Sensitivity (tornado)	DVH plots → 'Sensitivity (tornado)' button	Shows WHICH single parameter drives the most NTCP uncertainty.
4D dose accumulation	DVH import → 'Accumulate DVHs (4D)' button	Combines 2+ DVHs of the same structure (breathing phases, or primary+boost courses) into one accumulated DVH.
DICOM-RT import	DVH import → 'Load Eclipse/DICOM DVH' button	Reads DVHs directly from RTDOSE+RTSTRUCT files.
Pinnacle/Excel import	DVH import → 'Load DVH' button	Auto-detects and reads the native Pinnacle DVH export block format, or generic wide/long Excel layouts.
LKB parameter bank	Model/Endpoint parameters → 'Add from LKB parameter bank'; docs at Radiobiology → 'LKB parameter bank'	88 literature-sourced LKB parameter sets across 51 organs, with multiple citable options per endpoint and mandatory alpha/beta entry when a source didn't report one.
TCP parameter bank	Model/Endpoint parameters → 'Add from TCP parameter bank'; docs at Radiobiology → 'TCP parameter bank'	14 Target/Poisson TCP parameter sets across 9 tumour sites, with strict separation between total clonogen count (K) and per-cc density so the two are never silently confused.
Searchable dropdowns	Organ/Endpoint/Site fields in both parameter banks	Type any part of a name (case-insensitive) to filter instantly, e.g. "rect" jumps straight to "Rectum".
Export	Radiobiology → 'Export...'	Get DVH data, NTCP/TCP summaries, or the endpoint list out as CSV/JSON files.
Splash screen	Shown automatically on launch	Instant visual confirmation the app is starting, before the slower library imports happen.

Interpreting the confidence interval

The original paper explicitly lists the absence of confidence intervals as a limitation of BioSuite ("In its current state, BioSuite does not include any confidence interval calculations on NTCP and TCP..."). BioSuite-NG's Monte-Carlo CI fills that gap by sampling the model's own parameters (e.g. TD50 and m for LKB) from a normal distribution and recomputing NTCP thousands of times.

Important caveat: the ±10% standard deviation used for every parameter is currently a generic placeholder, not a value derived from the literature source of each specific parameter set. Treat the resulting interval as illustrative of how sensitive the result is to parameter uncertainty in general, not as a publication-ready statistical confidence interval, until the placeholder is replaced with real reported standard errors.

13. Troubleshooting

- **“No NTCP/TCP shows up in DVH plots”** — make sure you associated the structure's DVH with an endpoint in the DVH import tab (select the row, choose an endpoint, click 'Associate to DVH(s)'), and that the structure's row is selected/highlighted in the DVH plots tab's list.
- **“Compute 95% CI” says a model isn't wired up** — this currently covers LKB, RS, SMD and EUD NTCP models; TCP-endpoint CI is not yet implemented.
- **Chart looks squashed on a small window** — resize the window taller; the charts reserve fixed margins so axis labels are never clipped, but very small windows will still compress the plot area.
- **DICOM-RT import fails** — confirm both an RTDOSE and an RTSTRUCT file from the SAME plan/study are selected, and that the structure name you type matches the ROI name exactly as stored in the RTSTRUCT file.
- **“Add from LKB parameter bank” won't let me click OK** — the chosen parameter set's source did not report alpha/beta; type a value into the 'alpha/beta (Gy) to use' field first (see Section 5). This is deliberate: BioSuite-NG never invents a fractionation-correction value on your behalf.

14. References

Uzan J, Nahum AE. Radiobiologically guided optimisation of the prescription dose and fractionation scheme in radiotherapy using BioSuite. *Br J Radiol.* 2012;85(1017):1279–1286. doi:10.1259/bjr/20476567

See the project's README.md for the full list of model-parameter literature sources (Seppenwoolde et al., Rancati et al., Bentzen et al., Nahum et al., etc.) reproduced from the original paper's Tables 1 and 2.

The original BioSuite's own exercise sheet (“BIOSUITE EXERCISES SHEET”, CCO/J. Uzan, 2012) was also used directly to cross-check BioSuite-NG's engine: it confirmed the Relative Seriality model gives an NTCP close to the LKB model for the same lung endpoint (the exercise sheet's own worked example), and that the Simple Maximum Dose model is an essentially binary (0%/100%) sigmoid at realistic cGy-scale doses -- both reproduced exactly by BioSuite-NG's implementation.